

BOXI FU

Los Angeles, CA | boxifu@usc.edu | [Linkedin](#) | [Github](#) | [Portfolio](#) | [Scholar](#)

EDUCATION

- University of Southern California**, Los Angeles, CA May 2024
- *Master of Science in Computer Science* | **GPA: 3.9/4.0**
- University of California, Berkeley**, Berkeley, CA May 2022
- *Bachelor of Science in Computer Science*
 - *Bachelor of Science in Applied Mathematics* | **GPA: 3.8/4.0**

TECHNICAL SKILLS

Coding Languages: C#, C++, Python, Java, SQL; **Unity, Unreal**, Shiny App, OpenGL
AI/ML: LLM integration (Ollama, OpenAI API), prompt engineering, CV, Comfy UI, Stable Diffusion
Other Tools: Git, AWS, data pipelines, deployment workflows, state machines, event-driven architectures

EXPERIENCE

- Software Engineer at USC Institute for Creative Technologies** | Los Angeles, CA July 2022 — Present
- Developed and shipped 4+ interactive Unity-based games and public-facing installations, focusing on player interaction and real-time systems
 - Designed and implemented real-time interactive systems combining UI, vision, and audio inputs for player-facing applications
 - Integrated locally deployed AI models (emotion detection, facial tracking, dialogue systems)
 - Collaborated in small interdisciplinary teams to rapidly prototype and iterate gameplay and interaction systems

- Software Engineer and System Designer for *Monster Core* Indie Game** June 2021 — Dec 2021
- Designed and developed main game mechanics for *Monster Core*, a deck building game integrated with Pokemon style monster training system under an extraction framework, as one of three founding members of the indie game team
 - Participated in weekly meetings for regular updates, event planning, and group play testing weekly builds
 - Implemented combat system with Godot 5 engine with Trae and Codex AI assistive coding

- AR Research Intern at Innopeak Technologies (OPPO)** | Palo Alto, CA Jun 2021 — Dec 2021
- Built ML pipeline for 3D avatar generation from facial landmark data
 - Integrated datasets with 3D character systems
 - Processed motion capture and audio datasets

PROJECTS

- Monster Core*** (Godot 4, Indie Game) 2024 — Present
- Lead software engineer in 4–5 person team
 - Designed and implemented a modular combat system combining team-based switching, card-driven mechanics, and resource management
 - Implemented capture system with skill extraction, incorporated with local Gatcha pulling system for skill cards
 - Used Stable Diffusion for monster design iteration, and used Codex / Trae for AI-assisted development

- The 7th Patient*** (Unity, WebGL, Education, Narrative Diagnosis Game) June 2023 — Present
- Handled end-to-end system architecture as the sole engineer, from core logic to UI and deployment
 - Designed and implemented a state-driven system combining dialogue graphs, UI rendering, and real-time synchronization
 - Implemented adaptive hint system based on user behavior and system state
 - Deployed to 2,000+ users in constrained environments; optimized runtime performance under strict device limitations

- Desol*** (Unity, USC Games AGP) May 2025 — May 2026
- Software Engineer & Technical Artist
 - Developed a dynamic fog and skybox blending system to enhance environmental storytelling and player immersion
 - Implemented cel-shading pipeline
 - Created procedural sandfall particle effects and shaders for dynamic water container materials with render layer support
 - Worked on procedural terrain system

- Becoming Fei*** (Unity, Simulation, AI FPS Educational Game) Aug 2022 — Present
- Designed and implemented modular system architecture combining simulation logic, UI systems, and interaction workflows across multiple subsystems: Reconnaissance, Medic AI, & Equipment Maintenance
 - Developed subsystem interfaces for real-time data visualization and decision support, handling dynamic state updates and user interaction flows
 - Optimized deployment through asset streaming, compression, and performance tuning for constrained environments

PERSONAL INFORMATION

- Hobbies: Carillon, Hand Crafting, Graphic Design
- Languages: English (native), Mandarin Chinese (native)
- US Citizen